

Dark Matter — an Illusion

Why Galaxies Work Without Invisible Mass

*Fundamental Fractal Geometric
Field Theory (FFGFT) — Galactic Physics*

A Popular Science Overview

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<https://github.com/jpascher/T0-Time-Mass-Duality>

September 2026

Version 1.0 — Popular Science Edition

Keywords: Dark matter · rotation curves · galaxies · MOND · gravitational lensing · Bullet Cluster · FFGFT · inertia regime

Johann Pascher: *Dark Matter — an Illusion. Why Galaxies Work Without Invisible Mass. FFGFT.*

First edition, September 2026.

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Chapter 1

The Mystery of Flat Curves

It was a simple measurement with an uncomfortable answer.

In the 1970s, Vera Rubin observed something that physics has not let go of since: the stars at the edge of a galaxy move too fast. Far too fast. According to the law of gravity — the same law that keeps planets orbiting the Sun — stars far from the centre should move more slowly than those closer in. Just as Neptune orbits the Sun more slowly than Earth does.

But they do not.

The rotation curves of galaxies — the curve showing how fast stars at different distances from the centre orbit — flatten out at the edges. The velocity stays constant as far as one can measure. It is as if all planets in our solar system orbited the Sun at the same speed, no matter how far away they are.

The standard explanation is: there must be more mass than we see. An invisible, lightless mass — dark matter — must envelop the galaxies and strengthen the gravitational pull. Without it, the stars at the edge would fly away.

This explanation sounds reasonable. But it comes at a price: dark matter has never been directly detected. Not in particle accelerators, not in underground detectors, not through any other experiment. We postulate 26 % of the entire universe to consist of a substance we do not know, cannot see, and cannot grasp.

This book asks a different question: What if galaxies are not misbehaving — but our physics simply stops working at their edges?

FFGFT — Fundamental Fractal Geometric Field Theory — gives a precise answer. It needs no new particles. It needs no invisible mass. It needs only a single number: $\xi \approx \frac{4}{3} \times 10^{-4}$.

Chapter 2

What Newton Really Says — and Where He Stops

Newton's law of gravity is one of the most reliable formulas in physics. On Earth, in the solar system, in computing satellite trajectories — it works with extraordinary precision.

But it carries a hidden assumption: that mass and inertia always bear the same relationship to each other under all circumstances. Inertia is what prevents a body from changing its velocity — the greater the inertia, the more force is needed to accelerate it.

Newton wrote: $F = m \cdot a$. Force equals mass times acceleration. This seems self-evident. But what if this relationship only holds for large accelerations?

In everyday life, accelerations are large. A falling object on Earth experiences $g \approx 10 \text{ m/s}^2$. A satellite in Earth orbit perhaps a thousandth of that. But a star at the outermost edge of a galaxy experiences an acceleration of only $a \sim 10^{-10} \text{ m/s}^2$ — one ten-billionth of Earth's surface gravity.

Have we ever tested whether Newton's law holds in this regime?

The honest answer is: no. In the laboratory, in the solar system, on Earth — we have confirmed it thousands of times. In the galactic deep-field regime, we have never tested it directly.

FFGFT says: different physics applies there. Not because a new force appears, but because inertia itself becomes nonlinear.

Chapter 3

The Fundamental Relation: Time and Mass as Reciprocals

The core of FFGFT is a single formula:

$$\tilde{T} \cdot m = 1.$$

Intrinsic time and mass are reciprocals of each other. Where mass is large, time passes slowly. Where mass is small, time passes quickly. This is not a metaphor — it is a geometric property of the field.

From this relation it follows: a body with very small mass inhabits a time regime that differs fundamentally from that of a heavy body. And since inertia is connected to mass and time, inertia itself changes.

This sounds abstract. But it has a very concrete consequence for galaxies.

At the edge of a galaxy, the mass of visible matter is low, and the acceleration is very small. Both effects appear together: small mass and small acceleration. FFGFT says that in this regime, Newton's formula $F = ma$ no longer holds exactly. Inertia becomes smaller than Newton predicts — stars respond to gravity more sensitively than expected.

This is the effect we see in rotation curves — without dark matter.

Chapter 4

The Transition Scale a_0 — a Boundary from First Principles

For a theory to be scientific, it must predict precisely where an effect begins. FFGFT does this.

There is an acceleration scale a_0 below which the Newtonian regime ends and the galactic regime begins. This scale is not a free number to be fitted — it follows directly from the single parameter ξ of the theory:

$$a_0 = \frac{c^2 \xi^{10}}{4 \bar{\lambda}_e} \approx 1.03 \times 10^{-10} \text{ m/s}^2.$$

Here $\bar{\lambda}_e = \hbar/(m_e c)$ is the reduced Compton wavelength of the electron — a known, measurable quantity. The factor ξ^{10} carries the entire scale structure: ten powers of the fundamental constant connect the quantum scale of the electron to the scale of galactic acceleration.

This is remarkable. This value of a_0 had been known empirically in astrophysics for decades — Mordehai Milgrom determined it from observations in 1983, without being able to derive it. FFGFT derives it from first principles, with no free parameter.

Below a_0 , the modified force law applies:

$$a = \sqrt{a_N^2 + a_N a_0},$$

where a_N is the Newtonian acceleration from the visible mass. For large accelerations ($a_N \gg a_0$) this reduces to Newton: $a \rightarrow a_N$. For small accelerations ($a_N \ll a_0$):

$$a \rightarrow \sqrt{a_N a_0} \quad \Longrightarrow \quad v^4 = G M_b a_0.$$

This is the *baryonic Tully-Fisher relation* — one of the most robust empirical rules in galactic physics, observed for forty years, yet never derived from first principles by the standard model. FFGFT derives it.

Chapter 5

The Test: DDO 154 and the Milky Way

A theory must be measured against observations. Three galaxies of different types are particularly instructive.

DDO 154 — the cleanest test. This dwarf galaxy is gas-dominated: almost all of its visible mass consists of hydrogen gas, whose quantity is known very precisely from radio astronomy. There is little uncertainty about the visible mass — this makes DDO 154 the ideal test case.

The Newtonian prediction for the outer rotation velocity is about 18.5 km/s. The observation shows 47.0 km/s — more than twice as large.

The FFGFT prediction from the modified force law: 47.1 km/s.

The ratio of prediction to observation: 1.001 — agreement to one part per thousand.

NGC 3198 — a spiral galaxy. Here the baryonic mass is less precisely known (uncertainty $\sim 30\%$). The Newtonian expectation: 50.4 km/s, observed: 150 km/s. FFGFT predicts 121 km/s — a deviation of 18 %, fully within the mass uncertainty.

The Milky Way. Newton expects 122 km/s in the outer disc, measurements give around 220 km/s, FFGFT predicts 180 km/s. Again, the deviation lies within the stellar mass uncertainty.

Dark matter explains these curves too — but only by fitting a separate halo profile for each galaxy. Those are free parameters, one per galaxy. FFGFT has not a single free parameter: a_0 follows from ξ , which itself carries all of physics.

Chapter 6

The Bullet Cluster — the Seemingly Strongest Argument

Anyone who argues against dark matter eventually encounters the same objection: the Bullet Cluster.

In 2006, Douglas Clowe and colleagues published an observation considered decisive evidence for dark matter. Two galaxy clusters had collided. The hot gas — by far the largest visible mass component — was slowed by the collision and left behind. The galaxies themselves passed through.

From gravitational lensing measurements, the true location of the mass was determined: not at the gas, but at the galaxies. The gas was slowed; the mass was not. Therefore, a collision-free invisible mass must be present — dark matter.

This is a compelling argument. But FFGFT answers it without dark matter.

The modified force law acts on all mass — visible and interacting alike. What matters is the mass ratio between galaxies and gas in a cluster. The galaxies carry roughly ten times more mass than the gas. During the collision, the gas is slowed by radiation pressure and electromagnetic interaction — the galaxies are not.

In a finite-element simulation of the collision geometry (using the parameters of Clowe et al. 2006), FFGFT gives: the gravitational lensing peak lies 10 kpc from the galaxies and 240 kpc from the gas centre. This is exactly what is observed.

The Bullet Cluster is not evidence for dark matter. It is evidence that mass and gas respond differently in a collision. This has always been known — and in FFGFT it follows directly from the geometry.

Chapter 7

Dark Energy — a Construct of the Wrong Interpretation

Alongside dark matter, the standard model knows a second unknown: dark energy. It is supposed to drive the accelerated expansion of the universe and is said to account for 69 % of the entire universe.

Together, the two unknown components amount to 95 % of the universe. Only 5 % is ordinary matter that we know and understand.

FFGFT needs neither dark matter nor dark energy — but for different reasons.

Dark matter, as shown, is not a substance: it is a regime in which inertia becomes nonlinear. The effects are real — rotation curves are real, gravitational lensing is real — but they require no new ingredient.

Dark energy disappears for a different reason: there is no expansion to accelerate. In FFGFT the universe is static. The redshift of distant galaxies arises not because space is stretching, but because light loses energy along its path through the ξ -field:

$$1 + z = e^{\xi x}.$$

For small distances this is indistinguishable from a Doppler effect. For large distances it makes the same predictions as the standard model — because the data are degenerate: they cannot in principle distinguish between expansion and geometric path lengthening.

The cosmological constant Λ — the mathematical shorthand for dark energy — has no referent in a static universe. It is a book-keeping quantity of the expansion interpretation, not a physical object.

Chapter 8

What the Data Can Actually Decide

A fair question: if FFGFT can explain the same observations as the standard model — why switch?

The answer lies in the structure of the theories, not in individual data points.

The standard cosmological model (Λ CDM) describes cosmology with six free parameters, all empirically determined: Hubble constant, baryon density, dark matter density, spectral index, fluctuation amplitude, optical depth. The particle physics standard model adds 19 more. In total, over 25 numbers, none derived from first principles.

FFGFT has a single parameter: $\xi = \frac{4}{3} \times 10^{-4}$. From it follow:

- the gravitational constant G ,
- the Hubble constant $H_0 = 66.82 \text{ km/s/Mpc}$,
- the acceleration scale $a_0 = 1.03 \times 10^{-10} \text{ m/s}^2$,
- the CMB temperature $T_{\text{CMB}} = 2.725 \text{ K}$,
- the baryonic Tully-Fisher relation,
- the lepton mass hierarchy.

The same ξ anchors five different sectors of physics — particle physics, cosmology, galactic dynamics, gravitational lensing, quantum mechanics. That is not calibration: it is cross-anchoring.

The cosmological constant Λ , by contrast, works nowhere outside its own fit. That is the difference between a theory and a bookkeeping entry.

Where can one decide today? The classical pillars — supernovae, baryonic acoustic oscillations, CMB spectrum — are degenerate: both models produce the same mathematical curves. But there are observations that discriminate:

- **Wavelength dependence of redshift:** FFGFT predicts strictly achromatic redshift. Any demonstrated wavelength dependence would falsify the model.
- **Direct dark matter detection:** Any direct detection of a dark matter particle — in accelerators, detectors, or cosmic sources — would show that a new substance genuinely exists.
- **Euclid and DESI:** The new galaxy surveys will measure the growth of cosmic structure with unprecedented precision. FFGFT makes concrete predictions for the power spectrum without dark matter halo corrections.

Chapter 9

An Honest Assessment

FFGFT explains galaxy rotation curves without dark matter. It derives the acceleration scale a_0 from first principles. It passes the Bullet Cluster test. It explains why dark energy is not a physical object.

This is not a complete theory of gravity. Open questions remain:

The K3 forward proof — the mathematical derivation of the geometry from which the modified force law follows is structurally motivated but not yet fully proved.

14 % deviation at a_0 : The theoretically derived value lies about 14 % above the midpoint of the empirical range. However, the literature values for a_0 scatter between 1.1 and $1.3 \times 10^{-10} \text{ m/s}^2$ — the FFGFT value lies within this range.

Cluster physics: For large galaxy clusters with significant mass uncertainty, the predictions are less precise than for gas-dominated dwarf galaxies.

What is clear: assuming that 95 % of the universe consists of unknown substances that have never been directly detected is not a smaller assumption than assuming that inertia becomes nonlinear at very small accelerations.

The difference: the former assumption is not falsifiable — one can always add more dark matter when an observation fails to fit. The latter has a precise, distinguished scale a_0 that follows from the structure of the theory and manifests in observations.

Dark matter is not wrong. It might exist. But it is not necessary. And a physics that manages without it deserves serious scrutiny.

Chapter 10

What Comes Next

The next generation of observations will decide.

The ESA Euclid telescope is surveying billions of galaxies out to redshifts of $z \sim 2$. The James Webb Space Telescope is showing us galaxies in an early universe that, according to the standard model, should not yet exist — but fit comfortably in the static FFGFT universe. DESI is measuring baryonic acoustic oscillations with record precision.

For FFGFT, concrete predictions remain open:

Achromatic redshift — no wavelength dependence across the entire observable range — is a core test. Current precision observations already set tight limits; FFGFT is so far compatible.

The mass distribution in galaxy clusters without dark matter halos: FFGFT cluster profiles differ subtly from NFW profiles of standard dark matter models. The resolution of future lensing surveys could distinguish them.

The baryonic Tully-Fisher relation holds in FFGFT exactly — not as a statistical trend with scatter, but as a geometric consequence. Deviations would be a direct indication of dark matter as a genuine substance.

Whether dark matter exists or is an illusion of the wrong physics is one of the greatest open questions in science. It is decidable. The theory that answers it will not be found through belief — but through predictions that either prove themselves or fail.

FFGFT stands ready to be tested.

Sources and Foundation Documents

This book is based on the FFGFT corpus in the GitHub repository <https://github.com/jpascher/T0-Time-Mass-Duality>. The following documents are the primary sources:

- **Doc. 308** — The cosmic sector of FFGFT (redshift, dark matter, Bullet Cluster, light deflection)
- **Doc. 267** — Cosmological degeneracy: Λ CDM and FFGFT as two readings of the same observations
- **Doc. 309** — The scale-anchor problem (H_0 derivation, SI anchor, asymmetry of the models)
- **Doc. 025** — T0 cosmology (CMB, ξ -field effects, static universe)
- **Doc. 004** — T0 model overview (fundamental relations, field geometries)
- **Doc. 012** — Gravitational constant from FFGFT

All calculations in Chapters 4 and 5 were verified with the verification scripts `ffgft_308_p44_stufeA_a0_kette.py` and `ffgft_308_p44_stufeC_bullet_cluster.py`.